

The Three-Prime Running Least Common Multiple

A two-prime transcendence theorem, exact three-prime structure, and a conditional residue criterion

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ABSTRACT

For every pair of distinct primes, both versions of Erdős Problem #269 are transcendental: the original series, in which a reciprocal is repeated at every smooth integer, and the de-duplicated series, in which each running least common multiple contributes once. Ordering the two pure-power channels gives a Beatty word and reduces both sums to a Hecke–Mahler value. The proof uses the theorem of Loxton and van der Poorten in the modern form stated by Bugeaud and Laurent; it is an argument in this paper, not a Lean formalisation. Steve Fan posted the same reduction and conclusion before the first public release of this manuscript.

For three pairwise distinct primes p, q, r , we prove exact structure rather than irrationality. If $L(x)$ is the least common multiple of the $\{p, q, r\}$ -smooth numbers not exceeding x , then

$$L(x) = p^{\lfloor \log_p x \rfloor} q^{\lfloor \log_q x \rfloor} r^{\lfloor \log_r x \rfloor}.$$

This identity gives the logarithmic cells, jump ratios, finite height-fibre normal form, and a quadratic bound for short smooth shells. The reciprocal kernel is not finitely separable: for every order there is a nonsingular minor, uniformly in the third coordinate.

For $\{2, 3, 5\}$, grouping jumps between consecutive powers of two produces the exact radix alphabet $\{2, 6, 10, 30\}$. The literal shell tail satisfies an affine recurrence. Rationality places all of its normalised states on one rational lattice, while the integral branch is upward closed and rigid inside anchor intervals of subexponential width. These results identify the hard branch but do not exclude it.

The actual tail admits a quadratic bound in its logarithmic rank: if $n_a = \lfloor \log_2(2^a) \rfloor + \lfloor \log_3(2^a) \rfloor + \lfloor \log_5(2^a) \rfloor$, then $X_a \leq (n_a^2 + 8n_a + 18)/9$. This supplies the bound required by the carry argument. Rationality would therefore give a positive integral reduced carry, after cancellation of the smooth denominator factor. A denominator-dependent cofinal residue-window escape would contradict it. The actual-tail bridge is a paper proof; the finite residue contradiction and common-factor cancellation are checked in Lean. A reproducible integer scan finds an escaping window of length at most 18 for each of 106,666 tested denominator/start pairs.

The three-prime problem remains open. This criterion now has one unproved input: the actual $\{2, 3, 5\}$ radix and forcing words must satisfy cofinal residue-window escape.

**Authorship and AI use.* Will Cook built and directed the research infrastructure. AI agents produced the mathematical interpretations, proofs, prose, and software. Cook maintains the project and is responsible for its presentation; this is not a claim of personal mathematical verification or independent specialist review.

Proved cases, new structure, and limit

Two primes. Both versions of the series are transcendental for every pair of distinct primes, by reduction to a Hecke–Mahler value. **Three primes.** This paper proves the exact LCM cell structure and a quadratic bound for the actual $\{2, 3, 5\}$ tail. Together with the ordered-digit identity and smooth-factor cancellation, this reduces irrationality of that instance to the single cofinal residue-window escape condition in (9.5). **Open boundary.** A scan verifies escape only on 106,666 finite denominator/start pairs; the required unbounded-denominator and cofinal-start quantifiers remain unproved.

1 Introduction

Let P be a finite set of primes with $|P| \geq 2$, and let $a_1 < a_2 < \dots$ enumerate the positive integers all of whose prime factors lie in P . Erdős Problem #269 asks whether

$$\sum_{n \geq 1} \frac{1}{[a_1, \dots, a_n]}$$

is irrational, where $[a_1, \dots, a_n]$ is the least common multiple [1, p. 65][2, p. 106]. Bloom’s catalogue records the universal problem as open [4], but this note settles every two-prime instance; unresolved finite cases begin with $|P| = 3$.

In the primary 1988 source, Erdős states the infinite- P assertion as a simple exercise and presents persistence for a finite number of primes greater than one as a probable extension, not as a theorem; the current open status is supplied by the catalogue rather than inferred from that conjectural wording.

Three things are known and fix the shape of the question. The restriction $|P| \geq 2$ is necessary: for $P = \{p\}$ the enumeration is $a_n = p^{n-1}$, so $[a_1, \dots, a_n] = p^{n-1}$ and the sum is $p/(p-1)$, which is rational. For that reason we use the modern restriction $|P| \geq 2$; the 1974 letter itself writes “given primes $p_1, \dots, p_r$ ” without explicitly restricting r . For infinite P the sum is always irrational, which Erdős calls a simple exercise [2, p. 106]. And in a letter of 1 January 1973 he recorded that he could prove irrationality once duplicate running-LCM values are removed [3, p. 335]; the one-page letter states this result but does not include its proof.

That last remark is the one this note is closest to. The running least common multiple is not injective in n : it is constant along stretches of the enumeration and changes only at certain points. Removing duplicate summands means summing over the distinct values rather than over n . Section 4 proves that both this de-duplicated sum and the original repeated sum are transcendental when $|P| = 2$, by reducing them to a Hecke–Mahler value whose transcendence goes back to Loxton and van der Poorten [11, Theorem 8], quoted here in the modern form of Bugeaud and Laurent’s Theorem 1.1 [10, p. 61, Theorem 1.1].

This is an independent argument, not a recovery of the unprinted argument in the letter, and it is not the first public proof: Steve Fan posted the same factorisation, the same Hecke–Mahler reduction, and the same conclusion in the discussion thread of the problem’s page on 26 June 2026 [12], with follow-up remarks there extending the

argument to arbitrary coprime pairs. This manuscript was first released publicly on 22 July 2026, 26 days later, at commit a9d3ab8.

Sections 3 and 5 make the finite ingredients of the three-prime reindexing exact: they identify where the value is constant, by exactly what factor it changes when it changes, and give a finite rectangular-box fibre identity. Section 8 uses these ingredients to construct the actual infinite shell orbit.

Throughout, p, q, r are pairwise distinct primes, and a *pure power* is a power b^e of a single base; a *pure 3-power* is a power of 3, and similarly for the other bases. Call n *smooth* when $n = p^i q^j r^k$ for some $i, j, k \geq 0$; this is the [smooth lattice value](#).

For $x \geq 1$ write

$$L(x) = \text{lcm}\{n \leq x : n \text{ smooth}\}, \quad H(x) = p^{\lfloor \log_p x \rfloor} q^{\lfloor \log_q x \rfloor} r^{\lfloor \log_r x \rfloor},$$

the [running least common multiple](#) and the [pure-power height](#), where $\lfloor \log_b x \rfloor$ is the integer logarithm, the largest e with $b^e \leq x$. Since $a_1, \dots, a_n$ are exactly the smooth numbers up to a_n , we have $[a_1, \dots, a_n] = L(a_n)$, and the summands of the problem are the reciprocals of L along the enumeration. The reciprocal of the height at a smooth point is the [lattice kernel](#)

$$K(i, j, k) = \frac{1}{H(p^i q^j r^k)}.$$

For $b \in \{p, q, r\}$ we call the set of positive powers of b the *b-channel*.

Here “smooth” always means supported on the fixed prime set $\{p, q, r\}$. It is not the varying-bound notion counted by $\Psi(x, y)$ in the Dickman–Hildebrand theory [7, Theorem 1]; no smooth-number density asymptotic is used below. The relevant objects are the exact exponent lattice and its finite shells.

We treat $|P| = 3$ throughout, writing $P = \{p, q, r\}$, and the smallest instance is $\{2, 3, 5\}$. Theorem 9.1 determines exactly the alphabet of the dyadic compression of the *jump word*, that is, of the sequence of prime multipliers of L read in increasing order of the points where L increases (Section 3): the radix β_a of the block between 2^a and 2^{a+1} , meaning the product of the multipliers that occur in that block, takes one of the four values 2, 6, 10, 30 and no others.

Theorem 7.1 shows that the smallest two-by-two restriction of the kernel at $\{2, 3, 5\}$ has determinant $-1/15$ and hence rank two. Theorem 7.2 gives the full statement: non-singular minors exist in every order, uniformly in the remaining coordinate, and every finite exact separation $\sum_{\ell < d} f_\ell(i) G_\ell(j, k)$ is impossible. The mechanism is a small computation: $H(6) = 4 \cdot 3 \cdot 5 = 60$ and not 6, because the running least common multiple at a smooth cutoff already sees powers of the other primes that the cutoff itself does not contain.

The two-prime proof of Section 4 does not require such a separation.

The identification $L(x) = H(x)$ of Theorem 2.1 is what makes every statement about L below computable from three integer logarithms. In the unrestricted case that identification follows by iterating the prime-exponent maximum rule for least common multiples: $\text{lcm}(1, \dots, N) = \prod_{t \leq N} t^{\lfloor \log_t N \rfloor}$, the product being over the primes $t \leq N$ [6, Ex. 1.21(a), p. 22]. Chebyshev’s function is the corresponding prime-power sum, so equivalently $\log \text{lcm}(1, \dots, N) = \psi(N)$ [6, §4.2, p. 75]; Montgomery–Vaughan state this

exact identity directly in Exercise 6.2.7 [8, p. 183]. We record the P -restricted form because every later statement in the formal development is derived from it.

The line of argument developed below has four stages. First, identify L exactly as a product of three pure powers, so that every later statement is a statement about three integer logarithms. Second, compress its jump word into dyadic blocks, whose radix takes only the four values of Theorem 9.1.

Third, assume the sum rational with reduced denominator D , factor $D = D_{\text{sm}}B$ where every prime divisor of D_{sm} lies in $\{2, 3, 5\}$ and $\gcd(B, 30) = 1$, prove that the integral carry states, the integers the argument tracks from step to step (Section 9), share the factor D_{sm} , cancel it, and obtain a positive reduced carry d_n bounded by a denominator-dependent $K(B, n)$ and satisfying $d_{n+1} = b_n d_n - B m_n$. Fourth, find a stretch of consecutive steps over which the accumulated base and forcing put the residue of that carry above the bound, which is impossible.

The first two stages and the finite core of the fourth are proved and formalised here. The algebraic absorption-and-cancellation core of the third stage is also checked. The actual rationality-to-carry instantiation, including eventual divisibility by D_{sm} , is proved in this paper in Corollary 8.4, but is not yet formalised. The quadratic bound is proved in Theorem 8.3. The cofinal existence of windows in the fourth stage is the remaining hypothesis of the sufficient criterion in Section 10. The third and fourth stages follow a pattern standard in irrationality proofs; what is specific here is that the radix word driving the carry is the block alphabet of Theorem 9.1.

The statement of Problem #269 has been formalised before, as a conjecture with an unfilled proof, in the *Formal Conjectures* collection [5]. That is a formal statement of the question up to a harmless rational normalisation: its Nat-indexed series includes the empty-prefix least-common-multiple term. Its rational, irrational, and infinite-prime assertions all end in sorry. The declarations described below are propositions about the objects the question is posed over. Kovač and Tao [9] treat several irrationality problems of Erdős for series of unit fractions by elementary means; nothing from that work is used here.

Status. The original problem remains open. Statements marked as Lean-checked refer to the linked propositions accepted by the pinned kernel, with no sorry, added axioms, or unchecked evaluation. The note states the remaining mathematical obligations explicitly.

Companion system context. The [claim and trust boundary](#), [cold-clone route to proof authority](#), and [public contribution protocol](#) are described in sibling papers.

Results and boundary. The two transcendence theorems are arguments in this paper using the cited Bugeaud–Laurent theorem. The linked Lean declarations prove the exact three-prime running-lcm geometry, finite shell bounds and the rank-two kernel witness, the $\{2, 3, 5\}$ radix alphabet, and the finite residue contradiction. The formal denominator reduction and carry extinction retain their stated hypotheses. The paper shell argument supplies eventual smooth-factor divisibility and the required actual-carry bound. Co-final escape remains open. The dyadic-window scan covers 106,666 denominator/start pairs.

Structure

Section 2 identifies the running value. Section 3 develops its jump structure, and Section 4 uses that structure to prove transcendence of both two-prime sums. Section 5 gives the finite three-prime normal form, Section 6 bounds fibre multiplicity, and Section 7 proves arbitrary-order non-separability of the kernel. Section 8 constructs the actual infinite shell orbit, derives the all-scale rationality lattice, and analyses its integral branch. Section 9 determines the block alphabet, proves the conditional contradiction, and states the remaining unproved hypothesis. Section 10 collects the questions that remain. Linked phrases open the corresponding Lean declaration at the pinned source revision 4ab50e144847. The two transcendence theorems use the cited external theorem; arbitrary-order non-separability and the actual shell-orbit results are also paper proofs, not Lean-formalised endpoints.

Keywords. irrationality; least common multiple; smooth numbers; lattice sums; Lean 4.
MSC 2020. 11J72 (primary); 11A05, 11N25, 68V20 (secondary).

2 The running least common multiple as a product of pure powers

The smooth numbers up to x are indexed by the exponent triples (i, j, k) with $i \leq \lfloor \log_p x \rfloor$, $j \leq \lfloor \log_q x \rfloor$, $k \leq \lfloor \log_r x \rfloor$ and $p^i q^j r^k \leq x$: the coordinate bounds make the index set finite, and the last condition is the actual constraint. This is the [smooth prefix index set](#). Both conditions matter: the coordinate box is strictly larger than the prefix, since a product of three large pure powers can exceed x while each factor does not.

The identification below is routine, and its unrestricted analogue is classical, as recalled in the introduction; the short proof is given because every later statement is derived from it in the formal development.

Theorem 2.1 (the running least common multiple). *Let p, q, r be pairwise distinct primes and $x \geq 1$. Then $L(x) = H(x)$.*

Proof. For divisibility in one direction, every smooth $n \leq x$ has exponents bounded by the corresponding integer logarithms, so $n \mid H(x)$, and hence $L(x) \mid H(x)$. For the other, the three pure powers $p^{\lfloor \log_p x \rfloor}$, $q^{\lfloor \log_q x \rfloor}$ and $r^{\lfloor \log_r x \rfloor}$ are themselves smooth numbers not exceeding x , so each divides $L(x)$. Distinct primes have coprime powers, so their product divides $L(x)$ as well. The two divisibilities give equality. $\square$

Formalised as the [running-lcm identity](#), from the [divisibility into the height](#) and the three membership statements for the pure powers, namely the [first](#), [second](#), and [third](#) pure-power memberships.

The hypothesis that the primes are pairwise distinct is used exactly once, in the coprimality step, and it is not decorative: without it the three pure powers need not have coprime orders and their product need not divide the least common multiple. The identity is what makes every later statement about L computable from three integer logarithms.

At $(p, q, r) = (2, 3, 5)$ it reads $L(x) = 2^{\lfloor \log_2 x \rfloor} 3^{\lfloor \log_3 x \rfloor} 5^{\lfloor \log_5 x \rfloor}$, whose first ten values are

x	1	2	3	4	5	6	7	8	9	10
$L(x)$	1	2	6	12	60	60	60	120	360	360

So $L(10) = 8 \cdot 9 \cdot 5 = 360$, which is indeed the least common multiple of the smooth numbers 1, 2, 3, 4, 5, 6, 8, 9, 10, and $L(6) = 4 \cdot 3 \cdot 5 = 60$ rather than 6: the running value at a smooth cutoff already contains powers of the other two primes that the cutoff itself does not. These ten values also illustrate two of the statements of Section 3, since the value is constant on $\{5, 6, 7\}$ and on $\{9, 10\}$, and each change multiplies by a single prime, by 2 at $x = 2, 4, 8$, by 3 at $x = 3, 9$, and by 5 at $x = 5$.

Proposition 2.2. *For every $x \geq 1$, $H(x) \leq x^3$.*

Proof. Each of the three factors is a power of its base not exceeding x . □

Formalised as the [cubic majorant](#), which needs no primality. The exponent 3 is the number of generating primes, and the bound is the reason the reciprocal kernel is comparable to x^{-3} rather than to x^{-1} .

3 Constancy on logarithmic cells and the jump points

By Theorem 2.1 the running value depends on x only through the three integer logarithms. It is therefore constant wherever none of them changes and moves only where one of them does, and the next definition names the sets on which they are all constant, so that the next two theorems can say where the value stands still and by what factor it moves.

Say that x and y lie in the same *logarithmic cell* when $\lfloor \log_b x \rfloor = \lfloor \log_b y \rfloor$ for each of $b = p, q, r$: the [cell relation](#).

Theorem 3.1 (constancy on cells). *If $x, y \geq 1$ lie in the same logarithmic cell, then $L(x) = L(y)$. The same holds for the kernel at two smooth points whose values lie in one cell.*

Proof. Immediate from Theorem 2.1, since H depends on x only through the three integer logarithms. □

Formalised as the [cell constancy of the running value](#), the [cell constancy of the height](#), and the [cell constancy of the kernel](#). The running value therefore changes only when one of the three logarithms changes, and the next theorem says by exactly how much.

Theorem 3.2 (single-coordinate jump ratios). *Let $x, y \geq 1$. If $\lfloor \log_p y \rfloor = \lfloor \log_p x \rfloor + 1$ while the other two logarithms agree, then $L(y) = p L(x)$; and similarly with q or r in place of p .*

Proof. By Theorem 2.1 both sides are heights, and one factor of the height gains one in its exponent while the others are unchanged. □

Formalised as the [first](#), [second](#), and [third](#) coordinate steps, over the corresponding statements for the height alone, which need no primality: the [first height step](#), the [second](#), and the [third](#).

The jump points are therefore the pure prime powers, and they do not collide across channels.

Theorem 3.3 (jump count). *Let $n \geq 0$. The set of the first n positive powers of p , of q , and of r has exactly $3n$ elements, and adjoining the common origin 1 gives exactly $3n + 1$.*

Proof. The count is routine. Within one channel the powers $b, b^2, \dots, b^n$ are distinct because $b \geq 2$. Across two channels a common value would be a positive power of two distinct primes, which unique factorisation forbids. Finally 1 is not a positive power of any prime. $\square$

Formalised as the [positive jump count](#) and the [jump count with the origin](#), over the [channel cardinality](#), the [channel disjointness](#), and the [exclusion of the origin](#); the channels themselves are the [positive power sets](#). The exponent 0 is omitted from each channel because it is the shared initial value, which is why the origin is counted once rather than three times.

Two channels never meet, so at each positive pure power exactly one of the three logarithms advances, and by Theorem 3.2 the running value is multiplied there by the prime of that channel. Reading those multipliers in increasing order of the pure powers gives the *jump word* of L : one letter from $\{p, q, r\}$ for each positive pure power, recording which prime the value is multiplied by at that point. At $\{2, 3, 5\}$ the pure powers in increasing order are 2, 3, 4, 5, 8, 9, 16, 25, 27, 32, ..., so the jump word begins

$$2, 3, 2, 5, 2, 3, 2, 5, 3, 2, \dots$$

where the grouping is the one used in Section 9: each group ends at a power of two.

4 The two-prime sums are transcendental

The jump description gives a complete result in rank two. Temporarily let $P = \{p, q\}$ with $p < q$, and write

$$L_{p,q}(t) = p^{\lfloor \log_p t \rfloor} q^{\lfloor \log_q t \rfloor}.$$

The argument of Theorem 2.1, with one coordinate omitted, identifies this as the running least common multiple of the $\{p, q\}$ -smooth numbers up to t . Let

$$\mathcal{D}_{p,q} = 1 + \sum_{t \in \{p, p^2, \dots\} \cup \{q, q^2, \dots\}} \frac{1}{L_{p,q}(t)}.$$

Thus $\mathcal{D}_{p,q}$ retains the initial value and exactly one reciprocal for every later distinct running-LCM value.

Theorem 4.1 (two-prime de-duplicated transcendence). *For every pair of distinct primes p, q , the number $\mathcal{D}_{p,q}$ is transcendental.*

Proof. Interchanging the primes if necessary, assume $p < q$, and set

$$\theta = \frac{\log p}{\log q}, \quad x = \frac{1}{p}, \quad y = \frac{1}{q}, \quad m_n = \lfloor n\theta \rfloor, \quad \delta_n = m_{n+1} - m_n.$$

Here $0 < \theta < 1$, and θ is irrational: a rational relation would give $p^b = q^a$ for positive integers a, b . Consequently $\delta_n \in \{0, 1\}$.

The initial value together with the p -channel contributes

$$A = \sum_{n \geq 0} x^n y^{m_n}.$$

There is a q -power strictly between p^n and p^{n+1} exactly when $\delta_n = 1$; it is then q^{m_n+1} , and its post-jump reciprocal is $x^n y^{m_n+1}$. Hence the q -channel contributes

$$B = \sum_{n \geq 0} \delta_n x^n y^{m_n+1}, \quad \mathcal{D}_{p,q} = A + B.$$

All series here converge absolutely. Since $y^{m_{n+1}} - y^{m_n} = \delta_n y^{m_n} (y - 1)$, shifting the sum for A gives

$$A - 1 - xA = x(y - 1) \sum_{n \geq 0} \delta_n x^n y^{m_n}.$$

It follows that

$$\mathcal{D}_{p,q} = \frac{y - x}{x(y - 1)} A - \frac{y}{x(y - 1)}. \quad (1)$$

In the notation

$$F_\theta(x, y) = \sum_{n \geq 1} \sum_{k=1}^{\lfloor n\theta \rfloor} x^n y^k$$

for the Hecke–Mahler series, a finite geometric sum gives

$$A = \frac{1}{1 - x} - \frac{1 - y}{y} F_\theta(x, y). \quad (2)$$

Bugeaud and Laurent’s Theorem 1.1 states, in particular, that $F_\theta(\beta, \alpha)$ is transcendental when $\theta \in (0, 1)$ is irrational, α, β are nonzero algebraic numbers, $|\beta\alpha^\theta| < 1$, and $|\beta| < 1$ [10, p. 61, Theorem 1.1]; the $\rho = 0$ case used here goes back to Loxton and van der Poorten [11, Theorem 8]. We may take $(\beta, \alpha) = (x, y)$, because

$$|xy^\theta| = \frac{1}{p} \left(\frac{1}{q} \right)^{\log p / \log q} = \frac{1}{p^2} < 1.$$

Thus $F_\theta(x, y)$ is transcendental, and (2) makes A transcendental. Finally the coefficient of A in (1) is a nonzero algebraic number, since $x \neq y$; therefore $\mathcal{D}_{p,q}$ is transcendental. $\square$

The same value also controls the series before repeated running-LCM values are removed. Because each $\{p, q\}$ -smooth number is uniquely $p^i q^j$, that original series is

$$\mathcal{R}_{p,q} = \sum_{i,j \geq 0} \frac{1}{L_{p,q}(p^i q^j)}.$$

Theorem 4.2 (two-prime repeated-sum transcendence). *For every pair of distinct primes p, q , the number $\mathcal{R}_{p,q}$ is transcendental.*

Proof. Again assume $p < q$ and use $\theta, x, y, m_n, \delta_n, A, B$ from the preceding proof. At the smooth point $p^i q^j$,

$$L_{p,q}(p^i q^j) = p^{i+\lfloor j/\theta \rfloor} q^{j+m_i},$$

because $\log_p(p^i q^j) = i + j/\theta$ and $\log_q(p^i q^j) = j + i\theta$. Absolute convergence therefore permits the factorisation

$$\mathcal{R}_{p,q} = AC, \quad C = \sum_{j \geq 0} y^j x^{\lfloor j/\theta \rfloor}.$$

For each $j \geq 1$, put $n = \lfloor j/\theta \rfloor$. Irrationality of θ and $0 < \theta < 1$ give $m_n = j - 1$, $m_{n+1} = j$, and $\delta_n = 1$; conversely, every n with $\delta_n = 1$ arises from the unique $j = m_{n+1}$. Hence

$$C = 1 + B.$$

Equation (1), together with $\mathcal{D}_{p,q} = A + B$, gives

$$B = \frac{y(1-x)}{x(y-1)}A - \frac{y}{x(y-1)}.$$

Consequently

$$\mathcal{R}_{p,q} = \frac{y(1-x)}{x(y-1)}A^2 + \left(1 - \frac{y}{x(y-1)}\right)A.$$

The quadratic coefficient is a nonzero algebraic number. If $\mathcal{R}_{p,q}$ were algebraic, this display would make the transcendental number A a root of a nonzero polynomial over the algebraic numbers, a contradiction. $\square$

Thus both versions are settled, at the stronger level of transcendence, for $|P| = 2$. A third prime replaces the single Beatty boundary by a genuinely two-dimensional ordering problem, so neither theorem supplies a three-prime irrationality result.

5 The height-fibre normal form

Grouping a lattice sum by the value of the running least common multiple turns it into a sum over heights whose coefficients are multiplicities. We prove this exactly, on a finite rectangular box.

Write $\mathcal{B}(h_p, h_q, h_r)$ for the box of exponent triples with $i \leq h_p, j \leq h_q, k \leq h_r$, the [exponent box](#), and, for a height H , write $F(H)$ for the set of points of the box whose running height $H(p^i q^j r^k)$ equals H , the [height fibre](#). Distinct lattice points can carry the same height, and the coefficients of the normal form are exactly the sizes of these fibres.

Theorem 5.1 (finite normal form). *For every box $\mathcal{B} = \mathcal{B}(h_p, h_q, h_r)$,*

$$\sum_{(i,j,k) \in \mathcal{B}} K(i,j,k) = \sum_H \frac{\#F(H)}{H},$$

the outer sum ranging over the heights actually attained on $\mathcal{B}$.

Proof. The identity is a regrouping. Partition $\mathcal{B}$ into the fibres of the height map. On $F(H)$ every summand is $1/H$ by definition of the kernel, so the fibre contributes $\#F(H)/H$. $\square$

Formalised as the [height-fibre normal form](#), over the [fibre sum](#), with the height of a lattice point given by the [point height](#).

For a worked instance take $(p, q, r) = (2, 3, 5)$ and the box $\mathcal{B}(1, 1, 1)$, whose eight points carry the smooth values 1, 2, 3, 5, 6, 10, 15, 30 and the heights

$p^i q^j r^k$	1	2	3	5	6	10	15	30
H	1	2	6	60	60	360	360	10800

Six heights occur, two of them twice: the points 5 and 6 share the height 60, and 10 and 15 share the height 360. Theorem 5.1 here reads

$$1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{60} + \frac{1}{60} + \frac{1}{360} + \frac{1}{360} + \frac{1}{10800} = 1 + \frac{1}{2} + \frac{1}{6} + \frac{2}{60} + \frac{2}{360} + \frac{1}{10800} = \frac{18421}{10800}.$$

The two coefficients 2 carry the whole content of the regrouping on this box; where the heights are pairwise distinct the identity is a relabelling and nothing more.

Together with Theorem 3.1 this is the finite core of the ordered prime-power jump expansion: the value is constant on cells, the cells are indexed by heights, and the coefficient of a height is the number of lattice points it collects. The passage to the infinite sum, and the explicit ordering of the pure powers that would make the expansion a series in the jumps, are not proved here. Theorem 5.1 is an identity between two finite sums, and it is stated over the full box rather than the smooth prefix, so it is not a statement about L at a cutoff.

The same module records a one-step map $\tau(b, d, s) = b(s - d)$, the [variable-base tail step](#), with the rewriting [that names its expanded form](#). No orbit of this map is analysed.

6 A quadratic bound for smooth exponent shells

A tail estimate needs to know how many lattice points can share a short multiplicative interval. Fix a box $\mathcal{B}(h_p, h_q, h_r)$ and an interval $[\lambda, \eta)$, and write $\mathcal{S}$ for the set of exponent triples of that box whose smooth value lies in that interval, the [smooth exponent shell](#). Every bound below assumes the interval short in the multiplicative sense, meaning that η is at most a stated multiple of λ .

Informally, the next lemma says that an interval whose right endpoint is at most b times its left endpoint contains at most one of the numbers $b^a w$.

Lemma 6.1 (uniqueness in a short interval). *Let $b \geq 1$ and suppose $\eta \leq b\lambda$. If $b^a w$ and $b^{a'} w$ both lie in $[\lambda, \eta)$, then $a = a'$.*

Proof. If $a < a'$ then $\eta \leq b\lambda \leq b \cdot b^a w = b^{a+1} w \leq b^{a'} w < \eta$, which is impossible; the case $a > a'$ is symmetric. $\square$

Formalised as the [short-interval uniqueness](#). The hypothesis is that the interval has multiplicative width at most b .

Proposition 6.2. *Let $\mathcal{S}$ be the shell of the box $\mathcal{B}(h_p, h_q, h_r)$ in $[\lambda, \eta]$. If $\eta \leq r\lambda$ then $\#\mathcal{S} \leq (h_p + 1)(h_q + 1)$; if $\eta \leq p\lambda$ then $\#\mathcal{S} \leq (h_q + 1)(h_r + 1)$.*

Proof. Suppose $\eta \leq r\lambda$. If two triples of $\mathcal{S}$ agree in their first two coordinates, Lemma 6.1 applied with $b = r$ and $w = p^i q^j$ forces their third coordinates to agree as well. So the projection forgetting the third coordinate is injective on $\mathcal{S}$, and its image lies in a rectangle with $(h_p + 1)(h_q + 1)$ points. The case $\eta \leq p\lambda$ is the same with the first coordinate projected away. $\square$

Formalised as the [third-coordinate projection](#) and the [first-coordinate projection](#).

Theorem 6.3 (quadratic multiplicity bound). *Let $\mathcal{S}$ be the shell of the box $\mathcal{B}(h_p, h_q, h_r)$ in $[\lambda, \eta]$. Suppose $\eta \leq r\lambda$ and $h_p \leq h_q \leq h_r$ with $h_p + h_q + h_r = j$. Then*

$$9\#\mathcal{S} \leq (j + 3)^2.$$

Proof. By Proposition 6.2, $\#\mathcal{S} \leq (h_p + 1)(h_q + 1)$; under the sorting hypothesis the two surviving coordinates are the two smallest. It therefore suffices to prove that $a \leq b \leq c$ with $a + b + c = j$ gives $9(a + 1)(b + 1) \leq (j + 3)^2$. From $a \leq b \leq c$ we get $a + 2b \leq j$, so it is enough that $9(a + 1)(b + 1) \leq (a + 2b + 3)^2$; writing $b = a + d$ with $d \geq 0$, the difference of the two sides is $d(3a + 4d + 3) \geq 0$. $\square$

Formalised as the [quadratic shell bound](#), over the [sorted quadratic estimate](#). The constant 9 is the square of the number of generating primes and appears because the bound is the arithmetic–geometric comparison for a sum of three sorted coordinates; sorting is a hypothesis, not a normalisation, since the shell itself is not symmetric in the three bases. The last inequality of the proof is an equality when $h_p = h_q = h_r$, both sides then being $9(h_p + 1)^2$, so no constant larger than 9 survives that step.

The bound is uniform in λ and η subject to the width condition, and it is stated for the actual filtered shell rather than for a lattice model of it. It is an input to a tail estimate and is not itself one: no series is bounded here. The estimate is elementary and uses no analytic input on the distribution of smooth numbers, only the projection of Proposition 6.2.

7 Non-separability of the three-prime kernel

One might hope to write the three-prime kernel as $f(i)g(j)h(k)$ and so reduce the problem to one-dimensional criteria. Such a factorisation would force the value at $(1, 1, 0)$ to be determined by the values at $(0, 0, 0)$, $(1, 0, 0)$ and $(0, 1, 0)$, since the four values would then satisfy $K(0, 0, 0) K(1, 1, 0) = K(1, 0, 0) K(0, 1, 0)$. Theorem 7.1 computes those four values at $\{2, 3, 5\}$ and finds that they do not: the failure is exact, and it occurs on the smallest rectangle on which it could occur.

Theorem 7.1 (non-separability at $\{2, 3, 5\}$). *With $(p, q, r) = (2, 3, 5)$,*

$$K(0, 0, 0) K(1, 1, 0) - K(1, 0, 0) K(0, 1, 0) = -\frac{1}{15} \neq 0.$$

Consequently this two-by-two kernel restriction has rank two over $\mathbb{Q}$.

Proof. The four values are $K(0,0,0) = 1$, $K(1,0,0) = 1/2$, $K(0,1,0) = 1/6$ and $K(1,1,0) = 1/60$, computed from $H(1) = 1$, $H(2) = 2$, $H(3) = 2 \cdot 3 = 6$ and $H(6) = 4 \cdot 3 \cdot 5 = 60$. Hence the determinant is $1/60 - 1/12 = -1/15$, and a two-by-two matrix with nonzero determinant has rank two. $\square$

Formalised as the [non-separation witness](#), over the four exact values, the [origin](#), [value at two](#), [value at three](#), and [value at six](#). The exact determinant calculation is the [rank-two certificate](#). The height at 6 is 60 rather than 6 because the maximal pure powers below 6 are 4, 3 and 5: the running least common multiple at a smooth cutoff sees powers of the other primes that the cutoff itself does not contain. That is the mechanism behind the non-separation.

A nonzero two-by-two minor rules out writing the kernel as $f(i)g(j)h(k)$ on this box. The next theorem gives the general obstruction.

Theorem 7.2 (arbitrary-order non-separability). *Let p, q, r be primes with $p \neq q$, $p \neq r$ and $q \neq r$. For every $n \geq 0$ there are injective maps $I, J : \{0, \dots, n-1\} \rightarrow \mathbb{N}$ such that, for every $k \geq 0$,*

$$\det(K(I(a), J(b), k))_{0 \leq a, b < n} \neq 0.$$

Consequently, for no finite d do there exist rational-valued functions $f_\ell(i)$ and $G_\ell(j, k)$, $0 \leq \ell < d$, satisfying

$$K(i, j, k) = \sum_{\ell < d} f_\ell(i) G_\ell(j, k) \quad \text{for all } i, j, k.$$

Proof. Put $\alpha = \log_r p$, $\beta = \log_r q$, $x_i = \{i\alpha\}$ and $y_j = \{j\beta\}$. The height exponents are

$$\begin{aligned} v_p(H(p^i q^j r^k)) &= i + \lfloor j \log_p q + k \log_p r \rfloor, \\ v_q(H(p^i q^j r^k)) &= j + \lfloor i \log_q p + k \log_q r \rfloor, \\ v_r(H(p^i q^j r^k)) &= k + \lfloor i\alpha \rfloor + \lfloor j\beta \rfloor + \lfloor x_i + y_j \rfloor. \end{aligned}$$

Thus $K(i, j, k) = R_i(k) C_j(k) r^{-\lfloor x_i + y_j \rfloor}$ for nonzero rational row and column factors. In particular, the remaining carry matrix is independent of k .

The ratios α, β are irrational: a rational ratio would give an equality of positive powers of distinct primes. Their fractional-part orbits are therefore dense in $(0, 1)$. For $n > 0$ choose $0 < x_{I(0)} < \dots < x_{I(n-1)} < 1$. Choose $y_{J(0)}$ between $1 - x_{I(0)}$ and 1, and for $b > 0$ choose $y_{J(b)}$ between $1 - x_{I(b)}$ and $1 - x_{I(b-1)}$. The intervals are disjoint, so both index maps are injective. The resulting carry matrix has entry r^{-1} on and below the diagonal and 1 above it. Subtracting each preceding row from its successor, starting with the last row, gives determinant $r^{-1}(r^{-1} - 1)^{n-1} \neq 0$. The removed factors are nonzero for every k , proving the asserted uniformity. For $n = 0$ the empty determinant is 1. A representation with d summands would factor every $(d+1) \times (d+1)$ matrix through a d -dimensional space and force its determinant to vanish, contradicting the first assertion. $\square$

The exact $-1/15$ computation in Theorem 7.1 is the smallest visible witness. Theorem 7.2 excludes every finite exact separation of the displayed form, but it is still not an independence, irrationality or transcendence statement.

The linked declarations check the exact running-LCM identity, logarithmic-cell constancy, the $3n$ positive-jump count, the finite height-fibre normal form, the quadratic shell-multiplicity bound, the determinant $-1/15$. Arbitrary-order non-separability is the paper proof above; no linked Lean declaration currently formalises its dense-rotation construction or its arbitrary-size minors.

8 The actual shell orbit and its integral branch

We now pass from finite kernel structure to the literal infinite $\{2, 3, 5\}$ shell sum. Let s_a be the total reciprocal running-height mass of the smooth exponent triples in $[2^a, 2^{a+1})$, put

$$U_a = \sum_{n \geq 0} s_{a+n}, \quad X_a = \frac{H(2^a)}{2} U_a,$$

and let $b_a = \beta_a \in \{2, 6, 10, 30\}$ be the ordered block radix. For $a \geq 1$ let $d_a = m_a^{235}$ be the finite ordered digit defined in (9.1); set $d_0 = 1$. The following proof identifies these digits with the literal shell masses.

Theorem 8.1 (actual shell-orbit dichotomy). *The series $\sum_{a \geq 0} s_a$ is summable, and for every $a \geq 0$,*

$$X_{a+1} = b_a X_a - d_a.$$

Moreover, either $X_a \in \mathbb{Z}$ for some a , or for every a_0 there is $a \geq a_0$ such that

$$|X_a - z| \geq \frac{1}{31} \quad \text{for every } z \in \mathbb{Z}.$$

Proof. Every exponent of 3 or 5 in the shell is at most a , and for each pair of such exponents there is at most one exponent of 2, by the short-interval uniqueness lemma. There are therefore at most $(a+1)^2$ points. Each height is at least 2^a , giving

$$0 < s_a \leq (a+1)^2 2^{-a}.$$

The upper bound is summable; the strict lower bound follows from the point 2^a . In particular, all tail splittings below are justified.

Here is the finite identity that connects the ordered digit to this tail. Write the internal jumps in increasing order as $t_\ell = p_\ell^{e_\ell}$, $1 \leq \ell \leq m$, and set $t_0 = 2^a$, $t_{m+1} = 2^{a+1}$. Let $N(t)$ count smooth positive integers strictly below t , and put $n_\ell = N(t_\ell)$. The height on $[t_\ell, t_{\ell+1})$ is $H(2^a) \prod_{h \leq \ell} p_h$. The terminal jump has factor 2, so

$$H(2^{a+1}) = b_a H(2^a), \quad \frac{H(2^{a+1})}{2} s_a = \sum_{\ell=0}^m (n_{\ell+1} - n_\ell) \prod_{h > \ell} p_h.$$

Finite summation by parts rewrites the last expression as

$$n_{m+1} - n_0 + \sum_{\ell=1}^m (p_\ell - 1) \left(\prod_{h > \ell} p_h \right) (n_\ell - n_0).$$

With A_p, C_p as in (9.1), counting by the exponent of p gives $N(p^e) = C_p(e)$. Hence $n_{m+1} - n_0 = A_2(a+1)$ and the displayed expression is exactly $m_a^{235} = d_a$. At $a = 0$ the sole shell point is 1, giving $d_0 = 1$ directly. Thus d_a is a positive integer and

$$d_a = \frac{H(2^{a+1})}{2} s_a. \quad (3)$$

Splitting $U_a = s_a + U_{a+1}$ now gives $X_{a+1} = b_a X_a - d_a$. The bounded-radix alternative, whose abstract affine-orbit theorem is formalised as [the integral-state or cofinal-distance alternative](#), applies since $2 \leq b_a \leq 30$. The shell convergence and the identification (3) are proved here, not in that Lean declaration. $\square$

The same finite clearing works simultaneously at all scales. Write $h_a = H(2^a)/2$, which is an integer for $a \geq 1$.

Theorem 8.2 (all-scale rationality lattice). *Every finite shell window clears at its upper normaliser: for all $u, v \geq 0$ there is $m \in \mathbb{N}$ such that*

$$h_{u+v} \sum_{i=0}^{v-1} s_{u+i} = m.$$

If $q > 0$ and $U_1 = p/q$ for integers p, q , then

$$qX_a \in \mathbb{Z} \quad (a \geq 1).$$

Consequently there are $i < j$ and $z \in \mathbb{Z}$ with

$$X_{1+j} - X_{1+i} = z.$$

Proof. If $v = 0$, the window sum is zero, including at $u = 0$. Otherwise put $b = u + v \geq 1$. For every integer $1 \leq x < 2^b$, the 2-exponent in $H(x)$ is at most $b - 1$, while its 3- and 5-exponents are at most those in $H(2^b)$. Thus

$$H(x) \mid H(2^b)/2 = h_b.$$

The strict upper endpoint is what permits division by two. Every summand in the finite window therefore clears at h_b , proving the first assertion. Splitting U_1 into that finite window and the tail at a , multiplying by qh_a , and using the clearing identity proves $qX_a \in \mathbb{Z}$. Among $q + 1$ such integers two have the same residue modulo q ; their corresponding states differ by an integer. $\square$

Theorem 8.3 (quadratic bound for the actual tail). *Put*

$$n_a = a + \lfloor \log_3(2^a) \rfloor + \lfloor \log_5(2^a) \rfloor, \quad v_a = \#\{p^e < 2^{a+1} : p \in \{2, 3, 5\}, e \geq 1\}.$$

For every $a \geq 0$,

$$0 < X_a \leq \frac{n_a^2 + 8n_a + 18}{9} < \frac{v_a^2 + 10v_a + 27}{9}.$$

Proof. Partition the smooth integers $x \geq t = 2^a$ by their logarithmic height vector $(A, B, C) = (\lfloor \log_2 x \rfloor, \lfloor \log_3 x \rfloor, \lfloor \log_5 x \rfloor)$. A nonempty cell is the interval

$$[L, U), \quad L = \max(2^A, 3^B, 5^C), \quad U = \min(2^{A+1}, 3^{B+1}, 5^{C+1}).$$

It has $U \leq 2^{A+1} \leq 2L$. Fixing the exponents of 3 and 5 therefore leaves at most one exponent of 2 among its smooth integers. Since $A \geq B \geq C$, its multiplicity is at most

$$(B+1)(C+1) \leq \frac{(A+B+C+3)^2}{9},$$

by the sorted quadratic inequality used in Theorem 6.3. Here sorting follows from the ordered bases, rather than being an extra hypothesis. Unique prime factorisation identifies these exponent triples with distinct smooth integers.

There is at most one nonempty cell of each rank $n = A+B+C$: as the cutoff increases, all three logarithms are nondecreasing, and equality of their sum forces equality of their vector. Every cell above t has each coordinate at least its value at t . A cell of rank $n_a + k$ consequently has height at least $H(t)2^k$, since each of the k extra prime factors is at least 2. Nonnegative summation over cells, allowing empty ranks, gives

$$X_a \leq \frac{1}{18} \sum_{k \geq 0} \frac{(n_a + k + 3)^2}{2^k} = \frac{n_a^2 + 8n_a + 18}{9}.$$

The last equality uses the geometric moments $\sum 2^{-k} = 2$, $\sum k2^{-k} = 2$ and $\sum k^2 2^{-k} = 6$. Finally $\nu_a \geq n_a$, because every positive pure power at most 2^a is strictly below 2^{a+1} . Thus the last inequality is strict, with at least $(2n_a + 9)/9$ to spare. Positivity was proved above. $\square$

The finite projection and sorted quadratic inequality are checked Lean ingredients. Grouping the actual infinite tail by its successive height cells and summing the majorant is the paper argument above; it is not yet a formalised actual-tail theorem. The gain is that each additional cell rank costs a geometric factor while its multiplicity grows only quadratically.

Corollary 8.4 (eventual smooth-factor cancellation). *Suppose the original repeated series is $S = P/D$, with $D > 0$, and write $D = D_{\text{sm}}B$, where $D_{\text{sm}} = 2^u 3^v 5^w$ and $\gcd(B, 30) = 1$. Choose $a_D \geq 1$ so that $2^{a_D} \geq \max(2^{u+1}, 3^v, 5^w)$. Then for every $a \geq a_D$, DX_a is a positive integer divisible by D_{sm} . Consequently BX_a is a positive integer and*

$$BX_{a+1} = b_a BX_a - B d_a.$$

Proof. The original series is $S = \sum_{a \geq 0} s_a$, by unique prime factorisation and summability. Clearing its finite prefix at h_a gives $DX_a = h_a P - Dm$ for an integer m . Our choice of onset ensures $D_{\text{sm}} \mid h_a$: the extra dyadic factor accounts for $h_a = H(2^a)/2$. Thus $D_{\text{sm}} \mid DX_a$. Positivity and the recurrence follow from Theorem 8.1. $\square$

Thus pairwise incongruence of the actual X_a modulo one would prove irrationality. The source does not establish it: rationality predicts a collision rather than contradicting any known orbit theorem.

The integral branch in Theorem 8.1 has a further exact structure.

Theorem 8.5 (pinning and rigidity of the integral branch). *For every $a \geq 0$,*

$$X_a = \frac{d_a}{b_a} + \frac{X_{a+1}}{b_a}, \quad X_a > 0.$$

If X_a is integral, then X_n is integral for every $n \geq a$.

More generally, fix A , a positive width function w , and a real orbit $(y_n)_{n \geq A}$ satisfying $y_{n+1} = b_n y_n - d_n$. Suppose that for every $n \geq A$, both y_n and X_n lie in

$$\left(\frac{d_n}{b_n}, \frac{d_n}{b_n} + w(n) \right],$$

and that for every $\varepsilon > 0$ there is k_0 such that $w(A+k)/2^k < \varepsilon$ for all $k \geq k_0$. Then $y_A = X_A$.

Proof. The first identity is the recurrence solved for X_a ; positivity follows because every shell contains the dyadic point 2^a . Integer coefficients in the forward recurrence give upward closure. For the final assertion, the difference of the two orbits is multiplied by $b_n \geq 2$ at each step, whereas the common window bounds its absolute value at time $A+k$ by $w(A+k)$. Hence

$$2^k |y_A - X_A| \leq w(A+k).$$

The stated decay forces $y_A = X_A$. □

This theorem turns indefinite survival of an integral seed inside the exact windows into equality with the genuine infinite tail. It does not prove that no seed survives. The actual-orbit results in this section are paper proofs above. Lean checks the finite height, shell-count and radix ingredients and the abstract bounded-radix alternative; the actual infinite shell bridge, all-scale lattice, smooth-factor cancellation for this orbit, the actual quadratic tail bound and window-rigidity theorem are not yet formalised here.

9 Dyadic blocks and a conditional carry contradiction

The section makes three moves and leaves one hypothesis standing. We first compress the jump word of Section 3 into the blocks cut out by consecutive powers of two, and show that the multiplier of a block takes only four values (Theorem 9.1). We then record what can be cancelled from a hypothetical denominator, and exactly where that cancellation is still conditional (Theorem 9.2). Finally we leave the smooth numbers behind and argue with integer sequences alone: for a multiplier coprime to 30, no positive sequence obeying the cleared recurrence can stay inside its bound once a certain residue condition holds arbitrarily far out (Theorem 9.4). That residue condition, condition (E) below, remains unproved; the closing subsection reports a finite scan.

9.1 The four-element block alphabet

Take $(p, q, r) = (2, 3, 5)$ and compress the jump word of Section 3 between consecutive powers of two: a block starts just after 2^a , includes every pure 3- or 5-power strictly between 2^a and 2^{a+1} , and ends with the jump at 2^{a+1} . A channel cannot occur twice inside one block. Indeed, if

$$2^a < b^e, b^f < 2^{a+1} \quad (b \geq 2),$$

then the ratio between the interval endpoints is $2 \leq b$, so strict monotonicity of the powers forces $e = f$. This is the [checked internal-power uniqueness lemma](#). Each block therefore contributes at most one letter from each of the 3- and 5-channels, and exactly one letter 2 at its right end.

Let β_a , the *block radix*, be the product of the terminal dyadic factor 2, a factor 3 when the block contains an internal 3-power, and a factor 5 when it contains an internal 5-power; equivalently, β_a is the product of the letters of the jump word lying in block a .

Theorem 9.1 (the dyadic block alphabet). *With $(p, q, r) = (2, 3, 5)$ and β_a as above, for every a ,*

$$\beta_a \in \{2, 6, 10, 30\}; \quad \text{in particular} \quad 2 \leq \beta_a \leq 30.$$

Proof. Internal-power uniqueness leaves two independent yes/no choices, one for the 3-channel and one for the 5-channel. Multiplying the terminal factor 2 by the selected channel factors gives exactly the displayed four cases. $\square$

All four letters already occur among the first five of the six blocks tabulated below:

a	$(2^a, 2^{a+1})$	internal 3-power	internal 5-power	β_a
0	(1, 2)	none	none	2
1	(2, 4)	3	none	6
2	(4, 8)	none	5	10
3	(8, 16)	9	none	6
4	(16, 32)	27	25	30
5	(32, 64)	none	none	2

Block 4 is the only one of these six carrying an internal power in both channels, and block 5 carries neither, so its radix falls back to the terminal factor alone. Multiplying the radices along a run of blocks gives the product of the jump-word letters over that run: for instance $\beta_1\beta_2 = 60$ is the product of the four multipliers at 3, 4, 5, 8.

The definition is the [dyadic block base](#); Lean checks both the [exact four-case alphabet](#) and the [bounded-radix consequence](#). The radix word is therefore constrained to four values, and no growth hypothesis on it is needed. Section 10 gives a literal finite formula for the corresponding block digit m_a^{235} , matching the integer-only checker. What is not yet checked in Lean is the theorem that identifies that digit, its tail and its sharp carry bound with the original repeated series under a rationality hypothesis.

9.2 The denominator reduction and its boundary

Let D be the reduced denominator of a hypothetical rational value, and write

$$D = D_{\text{sm}} B, \quad D_{\text{sm}} = 2^u 3^v 5^w, \quad \gcd(B, 30) = 1.$$

The coprimality condition below is therefore intended as the endpoint of a reduction from an arbitrary D , not as a restriction on which rational values are being considered.

The argument runs on a sequence of integers c_n , one for each step, called the *carry states*; they satisfy a recurrence $c_{n+1} = b_n c_n - D m_n$ of the shape displayed below, driven

by a radix word b_n and a forcing word m_n . The name is meant to suggest the integer left after clearing D from the n -th tail of the series. For the actual normalised shell tail, Corollary 8.4 supplies this reading and eventual divisibility in a paper proof. The formal statements below take the recurrence and common factor as hypotheses.

Every fixed $\{2, 3, 5\}$ -smooth factor divides the running height once the cutoff reaches that factor ([checked absorption](#)). If the denominator-cleared carry states c_n share the absorbed factor, so that $c_n = D_{\text{sm}}d_n$, Lean cancels it from

$$c_{n+1} = b_n c_n - D_{\text{sm}} B m_n$$

and obtains

$$d_{n+1} = b_n d_n - B m_n$$

([checked cancellation](#)). Positivity and the sharp denominator-dependent upper bound descend through the same positive factor ([checked bound transfer](#)), and the reduced carry inherits the exact window identity ([checked window transfer](#)).

Informally: provided the smooth part D_{sm} divides every carry state, it can be divided out of the recurrence, leaving the same four statements with a multiplier coprime to 30 in place of D . This remains a hypothesis of the formal consumer; Corollary 8.4 supplies it for the actual orbit after a_D .

Theorem 9.2 (conditional denominator reduction). *If the actual denominator-cleared carry for $D = D_{\text{sm}}B$ has the common-factor form $c_n = D_{\text{sm}}d_n$ with $D_{\text{sm}} > 0$, then its recurrence, positivity, bound, and window identity reduce to the same statements with multiplier B , where $\gcd(B, 30) = 1$.*

Height absorption alone does not imply divisibility of a carry. The additional identity $DX_a = h_a P - Dm$ is what proves it here. The formal consumer does not yet contain that infinite-tail bridge. The specified sharp upper bound for the actual orbit follows from Theorem 8.3; its infinite-tail proof is likewise not part of the formal consumer.

9.3 The window recurrence and the residue contradiction

The statements of this subsection are about integer sequences: they do not refer to L or to the smooth numbers. Applying them to the actual block data uses the paper identification and tail bound above.

Call a pair (ℓ, h) with $h > 0$ a *window*: the stretch of h consecutive steps beginning at index ℓ . To locate a carry at the end of a window without performing any division, one needs the multiplier and the forcing accumulated across the window, and those are the two sequences defined next. For an integer radix word b_n and forcing word m_n , define

$$\begin{aligned} W_{\ell,0} &= 1, & W_{\ell,h+1} &= b_{\ell+h} W_{\ell,h}, \\ F_{\ell,0} &= 0, & F_{\ell,h+1} &= b_{\ell+h} F_{\ell,h} + m_{\ell+h}. \end{aligned}$$

These are the [window base](#) and [window forcing](#). If an integral carry satisfies

$$d_{n+1} = b_n d_n - B m_n,$$

then induction gives the exact division-free identity

$$d_{\ell+h} = W_{\ell,h}d_{\ell} - BF_{\ell,h};$$

this is the [checked window identity](#).

For $C > 0$, let $\text{lpr}_C(x) \in \{1, \dots, C\}$ be the least positive representative of $x \bmod C$, with a zero residue represented by C . This is the [canonical representative](#). Lean checks both its [positive range](#) and its [congruence to the source integer](#). This convention matters: replacing $\text{lpr}_C(x)$ by $|x|$ would not be a modular statement.

Let $K(B, n)$ be a bound on the reduced carry at denominator B and step n . It enters as a parameter: the statements below hold for whichever function K is supplied. Theorem 8.3 supplies the particular K^{235} used for the actual series in Section 10.

Define *cofinal local-window escape* to mean that for every $B > 0$ coprime to 30 and every ℓ_0 , there are $\ell \geq \ell_0$ and $h > 0$ such that

$$C_{\ell,h} := |W_{\ell,h}| > 0 \quad \text{and} \quad K(B, \ell + h) < \text{lpr}_{C_{\ell,h}}(-BF_{\ell,h}). \quad (\text{E})$$

Both the quantifier over B and the dependence of K on B are part of the statement. Informally, and separately for each fixed $B > 0$ coprime to 30: however far out one starts, some window has a nonzero accumulated base, and the least positive residue of its accumulated forcing, weighted by $-B$ and taken modulo that base, exceeds the carry bound at the endpoint of the window. The exact unproved proposition is the [cofinal local-window escape condition](#).

The next statement is the finite core of the argument. Informally, it says that a positive integer of size at most K cannot be congruent modulo C to a number whose canonical positive residue exceeds K . The key point is the convention just fixed: because lpr_C represents a vanishing residue by C and not by 0, the proof has to treat that case separately, and the two branches conclude for different reasons.

Theorem 9.3 (the finite residue contradiction). *Let $C > 0$, $c > 0$, and $|c| \leq K$. If $c \equiv x \pmod{C}$ and $K < \text{lpr}_C(x)$, then the hypotheses are contradictory.*

Proof. The canonical representative lies in $\{1, \dots, C\}$. The inequalities put c strictly between 0 and C . If $x \equiv 0 \pmod{C}$, its positive representative is C while $c \bmod C = c \neq 0$. Otherwise both c and $\text{lpr}_C(x)$ are their own residues and congruence makes them equal, contradicting $|c| \leq K < \text{lpr}_C(x)$. $\square$

The natural-state version is the [finite contradiction](#); the integer carry version is the [integer form of the contradiction](#).

The same finite arithmetic gives the following exact classifier. If $C > 0$, $c > 0$, $|c| \leq C$, and $c \equiv x \pmod{C}$, then

$$\text{lpr}_C(x) = |c|.$$

This includes the endpoint correctly: the zero congruence class is represented by C , so $c = C$ gives $\text{lpr}_C(x) = C$, not zero. Lean checks this as [exact least-positive-residue classifier](#). Applying it to the actual series uses the integral carry and bound proved above; constructing arbitrarily late escaping windows remains unproved.

Theorem 9.4 (a conditional contradiction for bounded carries). *Let (b_n) , (m_n) and K be the radix word, the forcing word and the bound of this subsection, and assume the cofinal escape property (E). Fix $B > 0$ coprime to 30. There is no integral sequence d_n satisfying simultaneously*

$$d_{n+1} = b_n d_n - B m_n, \quad d_n > 0, \quad |d_n| \leq K(B, n) \quad (n \geq 0).$$

Proof. Choose one window supplied by (E). The checked window identity gives

$$d_{\ell+h} \equiv -BF_{\ell,h} \pmod{|W_{\ell,h}|}.$$

The endpoint state is positive and at most $K(B, \ell + h)$, whereas the canonical positive residue of the right-hand side is larger than this bound. Theorem 9.3 is the contradiction. $\square$

This is formalised as the [reduced-carry extinction theorem](#). Coprimality with 30 is used by (E) to select a window; once a window has been chosen, the finite contradiction does not use it. The formalisation carries the edge cases: $|W_{\ell,h}| = 0$ is excluded, a zero residue is represented by the full modulus, and positivity prevents the endpoint carry from being zero.

The later [absorbed-carry extinction theorem](#) packages the preceding cancellation and reduced-carry contradiction into one statement. Under the cofinal escape hypothesis, it takes a nonzero smooth factor, an exact factorisation $c_n = D_{\text{sm}} d_n$, the absorbed recurrence for c_n , and the positive short bound for d_n , and derives a contradiction. The paper shell argument supplies the factorisation, recurrence and specified bound; cofinal escape remains unproved.

9.4 A finite check of the escape condition

A *certificate* here is a finite tuple of integers recording one instance of the inequality in (E) for the data the checker constructs. It exists so that a reader can recheck the instance in a line, without running the checker and without touching the infinite part of the argument. The integer-only [dyadic-window checker](#) constructs the ordered pure-power jumps, the block bases, and the block digits from exact multiplicity counts. It reproduces the following certificates; the columns are denominator B , dyadic start a , window length h , endpoint jump index n , window base W , forcing F , least positive residue $R = \text{lpr}_W(-BF)$, and short bound K .

B	a	h	n	W	F	R	K
1	1	2	4	60	47	13	9
7	1	3	6	360	289	137	95
16	1	4	9	10800	8735	640	352

The first row reads as follows. The window starts at $a = 1$ and has length 2, so its base is the product of the two block radices, $W = \beta_1 \beta_2 = 6 \cdot 10 = 60$; the forcing accumulated over the window is $F = 47$; and $\text{lpr}_{60}(-47) = 13$, since $-47 + 60 = 13$, which exceeds the bound $K = 9$. The other two rows are read the same way, with $W = \beta_1 \beta_2 \beta_3 = 360$ and

$W = \beta_1\beta_2\beta_3\beta_4 = 10800$. The third row lies outside the domain of the escape condition, since $\gcd(16, 30) = 2$ while both (E) and Theorem 9.4 quantify only over $B > 0$ coprime to 30. It is displayed to illustrate the window arithmetic at greater depth, not as an instance of the escape condition.

A fresh scan over every $B \leq 1000$ coprime to 30 and every $100 \leq a \leq 500$ tested 106,666 pairs. In every case a window of length at most 18 made both $W > K$ and $\text{lpr}_W(-BF) > K$; the largest first successful length was 14. The computation uses integers only and is reproducible from the pinned checker. Neither the scan nor the three displayed certificates proves escape for unbounded B or for cofinally many starts.

10 Complements and further questions

Theorems 4.1 and 4.2 close both two-prime questions, at the stronger level of transcendence. The three-prime de-duplicated and repeated series are outside the one-dimensional Hecke–Mahler reduction used in those theorems and remain open. At three primes the exact unresolved statement is best separated from the now-proved actual-tail bridge and from the possible methods for excluding its integral branch.

10.1 The actual block data and its bounded carry

The checker already uses literal data, which we record so that the open problems have no unspecified forcing word. For $p \in \{2, 3, 5\}$ let q, r be the other two primes and put

$$A_p(e) = \#\{(i, j) \in \mathbb{N}^2 : q^i r^j < p^e\}, \quad C_p(e) = \sum_{u=1}^e A_p(u).$$

Let I_a be the increasing list of internal jumps (p, e) with $p \in \{3, 5\}$ and $2^a < p^e < 2^{a+1}$. For $(p, e) \in I_a$ let

$$\sigma_a(p, e) = \prod_{(q, f) \in I_a, p^e < q^f} q.$$

Then, for $a \geq 1$, the exact checker definitions are

$$\beta_a = 2 \prod_{(p, e) \in I_a} p, \quad m_a^{235} = A_2(a+1) + \sum_{(p, e) \in I_a} (p-1)\sigma_a(p, e)(C_p(e) - C_2(a)). \quad (9.1)$$

The first formula is the four-letter radix of Theorem 9.1; the second is the integer implemented by the pinned checker. If

$$\nu_a = \#\{p^e : p \in \{2, 3, 5\}, e \geq 1, p^e < 2^{a+1}\},$$

its exact tested short bound is

$$K^{235}(B, a) = \left\lfloor \frac{B(\nu_a^2 + 10\nu_a + 27)}{9} \right\rfloor. \quad (9.2)$$

Finally define the radix-digit tail

$$T_a = \sum_{j \geq a} \frac{m_j^{235}}{\beta_a \beta_{a+1} \cdots \beta_j}; \quad T_{a+1} = \beta_a T_a - m_a^{235}. \quad (9.3)$$

This T_a is the normalised shell state X_a , whereas U_a denotes the unscaled shell tail. Indeed, for $j \geq a$, the height-ratio and shell-digit identities give

$$\frac{m_j^{235}}{\beta_a \cdots \beta_j} = \frac{H(2^a)}{2} s_j.$$

Summing the nonnegative terms proves convergence and $T_a = X_a$ for $a \geq 1$. Corollary 8.4 therefore supplies the integral reduced carry BT_a after a_D . The actual-tail estimate also proves the specific integer bound required by the checker.

Corollary 10.1 (the actual bounded reduced carry). *Under the rationality hypothesis and after the onset a_D of Corollary 8.4, BT_a is an integer and*

$$0 < BT_a \leq K^{235}(B, a), \quad BT_{a+1} = \beta_a BT_a - Bm_a^{235}.$$

Proof. Integrality, positivity and the recurrence follow from Corollary 8.4 and $T_a = X_a$. Theorem 8.3 gives $BT_a < B(\nu_a^2 + 10\nu_a + 27)/9$; an integer below that real bound is at most its floor, which is $K^{235}(B, a)$. $\square$

The actual bridge is therefore complete in the paper argument: the literal digit, tail, integral recurrence, smooth-factor cancellation and required bound all refer to the same series. Formalising that bridge remains a separate task; the cofinal residue escape below is the remaining mathematical hypothesis of this method.

10.2 The intrinsic tail question

Problem 10.2 (exact nonintegrality of every reduced tail). For every $B \geq 1$ with $\gcd(B, 30) = 1$ and every $a \geq 1$, prove

$$BT_a \notin \mathbb{Z}. \tag{9.4}$$

This pointwise form is stronger-looking but cleaner than “cofinally nonintegral”: if BT_a is integral at one index, the recurrence

$$BT_{a+1} = \beta_a BT_a - Bm_a^{235}$$

makes it integral at every later index. Thus a direct solution of Problem 10.2, joined to Corollary 8.4, bypasses all residue-window machinery and does not require the sharp bound.

The bounded-radix theorem gives a useful exact reduction. Since $2 \leq \beta_a \leq 30$, any real affine tail orbit either hits an integer or is, cofinally often, at distance at least $1/31$ from every integer (checked). It does not exclude the integral branch; Problem 10.2 is exactly what must do so for the actual orbit.

10.3 A denominator-adaptive sufficient criterion

For the literal pair (m^{235}, K^{235}) , retain the window definitions of Section 9:

$$W_{\ell, h} = \prod_{j=0}^{h-1} \beta_{\ell+j}, \quad F_{\ell, 0} = 0, \quad F_{\ell, h+1} = \beta_{\ell+h} F_{\ell, h} + m_{\ell+h}^{235}.$$

Problem 10.3 (actual cofinal local-window escape). Prove the displayed quantifier order

$$\forall B \geq 1 \text{ (gcd}(B, 30) = 1), \forall a_0 \geq 1, \exists \ell \geq a_0 \exists h \geq 1 : \text{lpr}_{W_{\ell,h}}(-BF_{\ell,h}) > K^{235}(B, \ell + h). \quad (9.5)$$

Theorem 10.4 (irrationality from actual cofinal escape). *If (9.5) holds, then the original repeated $\{2, 3, 5\}$ running-LCM series is irrational.*

Proof. Suppose the series is rational. Corollary 10.1 supplies a denominator $B \geq 1$ coprime to 30 and, after a_D , positive integral states $z_a = BT_a \leq K^{235}(B, a)$ with the actual recurrence. Choose an escaping window beginning at $\ell \geq a_D$. Iterating gives

$$z_{\ell+h} = W_{\ell,h}z_\ell - BF_{\ell,h} \equiv -BF_{\ell,h} \pmod{W_{\ell,h}}.$$

Since the least positive residue is larger than $K^{235}(B, \ell + h)$, and is at most $W_{\ell,h}$, the positive endpoint state is strictly below $W_{\ell,h}$. It must equal that least positive residue, contradicting its upper bound. Thus rationality is impossible. $\square$

Every β_a is positive, so $W_{\ell,h} > 0$ is automatic. The cofinal quantifier is present for a substantive reason: the integral reduced carry and its bound are available after the denominator-dependent onset a_D , and (9.5) then supplies a window beyond that onset. Once such a window is chosen, the positive endpoint carry must equal the canonical residue exactly (checked), yet it lies in the possible carry set $\{1, \dots, K^{235}(B, \ell + h)\}$; the strict inequality excludes that set. This is a one-sided least-positive-residue statement, not two symmetric arcs around zero.

Two exact countermodels rule out tempting shortcuts. For $(W, F, B) = (6, 4, 1)$,

$$\text{lpr}_6(-4) = 2,$$

so the canonical residue need not be coprime to W . For $(W, F, B) = (60, 47, 37)$,

$$\text{lpr}_{60}(-37 \cdot 47) = 1,$$

so a fixed window has no denominator-independent positive residue lower bound. Therefore the window may genuinely depend on B ; neither a fixed finite computation nor a universal bounded-length guess addresses the unbounded denominator and cofinal-start quantifiers. The finite checker can reject a proposed sufficient condition, but supplies no evidence for either shortcut.

10.4 A two-dimensional analytic representation

For the de-duplicated series define

$$\mathcal{D}_{2,3,5} = 1 + \sum_{t \in \{2^n, 3^n, 5^n : n \geq 1\}} \frac{1}{2^{\lfloor \log_2 t \rfloor} 3^{\lfloor \log_3 t \rfloor} 5^{\lfloor \log_5 t \rfloor}}.$$

The dyadic coding is the joint rotation word

$$\delta_{3,a} = \lfloor (a+1)\theta_3 \rfloor - \lfloor a\theta_3 \rfloor, \quad \theta_3 = \frac{\log 2}{\log 3}, \quad \delta_{5,a} = \lfloor (a+1)\theta_5 \rfloor - \lfloor a\theta_5 \rfloor, \quad \theta_5 = \frac{\log 2}{\log 5},$$

with $\beta_a = 2^{\delta_{3,a}} 5^{\delta_{5,a}}$.

Problem 10.5 (function-faithful two-dimensional representation). Express $\mathcal{D}_{2,3,5}$ as a nonconstant algebraic combination of values of a specified two-dimensional Hecke–Mahler, cone-generating or multivariate Mahler function and verify every hypothesis of a published value theorem; or give a conditional theorem under an explicit logarithmic nondegeneracy hypothesis; or prove that the literal series has no representation in the specified finite-dimensional class.

Pairwise irrationality of θ_3 and θ_5 is not silently promoted to the orbit-closure or equidistribution hypothesis a two-dimensional theorem may need. The finite-observer formalisation isolates the precise faithfulness requirement: equality in a finite observer must imply equality after symbolic realisation, and a genuine finite-dimensional factorisation forces the realised symbolic span to be finite-dimensional ([residue–coboundary form](#), [symbolic realisation](#), [checked](#)). The same finite formulation keeps the carry residue and residue digit in their declared intervals ([carry interval](#), [digit interval](#)). No theorem here proves that the literal realised span is infinite, so a general finite separable decomposition is neither assumed nor declared excluded.

10.5 Further structural constraints

The radix alphabet itself extends without difficulty. For ordered primes $p_1 < \dots < p_s$, an interval (p_1^a, p_1^{a+1}) contains at most one power from each other channel, because consecutive p_i -powers have ratio $p_i > p_1$. Hence its block radix belongs to the 2^{s-1} -letter alphabet

$$\left\{ p_1 \prod_{i=2}^s p_i^{\varepsilon_i} : \varepsilon_i \in \{0, 1\} \right\}.$$

The nontrivial question is quantitative: prove effective recurrence or discrepancy for the actual four-letter $\{2, 6, 10, 30\}$ word, an asymptotic formula with an error term for the restricted two-dimensional shell counts that generate m_a^{235} , or the exact finite-separation rank of the literal three-prime kernel under a specified family of shifts.

The three-channel rigidity and carry-lift extinction theorems already exclude one proposed argument in four exact steps. Under channel surjectivity, ordinary block-nullity is equivalent to the perturbation being a coboundary of a channel potential ([potential classifier](#)). Zero perturbations on genuine $2 \rightarrow 3$ and $2 \rightarrow 5$ transitions then identify all three potential values and force the perturbation to vanish at every index. For an integral lift, that vanishing makes a nonzero initial lift error grow by the exact product of the successive bases; bases at least two make its absolute value at least 2^N , contradicting even a single index- N bound strictly below 2^N ([one-index extinction](#)). A single index therefore already suffices, and consequently no uniform bound on the lift error can hold either: under the same hypotheses a nonzero initial error is incompatible with any bound valid at every index ([uniform extinction](#)).

A separate four-state calculation reaches the obstruction earlier: four real states in $(0, 1)$ with unit-accuracy integral lifts and the two anchor equalities force the first complete 2-block sum to be 1, hence that block cannot be null ([first-block sum](#), [four-state obstruction](#)).

These lift conclusions remain conditional. The paper constructs the actual #269 orbit and its bounded integral carry under rationality, but it does not construct a faith-

ful lift with the two anchors or block-nullity required by these auxiliary statements. The affine alternative above may produce an arbitrary integral state, not necessarily zero; its cofinal $1/31$ separation is neither eventual nor positive-density and gives no unbounded distance. The four-state calculation supplies no cofinal windows, and `rational_of_scaledTail_integer` classifies no denominators. The actual carry supplies a weighted block defect instead, so any successful argument must use that weighted identity or construct a different faithful lift.

The obstructions identify three different limits. Arbitrary-order minors exclude a finite exact separation of the kernel; the bounded-radix alternative leaves integral tails untouched; and the residue examples show that coprimality alone gives no useful denominator-independent residue bound. The actual-tail estimate resolves the size issue: rationality supplies $1 \leq BT_a \leq K^{235}(B, a)$ after a denominator-dependent onset. For the window criterion, the remaining task is to force the literal word's residue outside this finite carry interval, with the window allowed to depend on B and to begin beyond that onset.

For the repeated $\{2, 3, 5\}$ series, the actual-tail bridge and bound are proved above. Theorem 10.4 leaves one sufficient arithmetic condition: cofinal escape for the literal digit word, with the window allowed to depend on the reduced denominator. That condition, or a direct exclusion of integral tails, remains unproved; Problem #269 stays open.

Statements and declarations

Artefact and data availability. The [pinned formal-source revision](#) contains the Lean sources, the fixed toolchain, the library manifest, and the exact dyadic-window checker used in the finite experiment. This manuscript provides navigation rather than proof authority.

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A Guide to the formal sources

Each linked phrase opens its Lean declaration at the pinned source revision 4ab50e144847. The running-LCM structure, residue arithmetic, and local-window bridge occupy separate modules. The following distinctions matter when reading them. The height statements hold for arbitrary bases, while the statements about L need the three primes to be distinct. The normal form of Section 5 is a finite identity over a rectangular box, not a convergence theorem. The actual shell convergence, arbitrary-order non-separability, rationality lattice and orbit pinning are paper proofs. The paper also supplies eventual divisibility of actual carries by the smooth denominator factor; the formal cancellation lemma assumes that divisibility. Finally, the formal cofinal-escape predicate is an unproved hypothesis of Theorem 9.4; its application to the actual $\{2, 3, 5\}$ word is not asserted.

References

- [1] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monogr. Enseign. Math. 28, Geneva, 1980, p. 65. For a possibly infinite prime set Q , the page states the infinite- Q irrationality and asks what happens for finite Q with more than one element.
- [2] P. Erdős, *On the irrationality of certain series: problems and results*, in A. Baker (ed.), *New Advances in Transcendence Theory*, Cambridge UP, 1988, pp. 102–109, doi:[10.1017/CBO9780511897184.009](https://doi.org/10.1017/CBO9780511897184.009).
- [3] P. Erdős, *Letter to the Editor* (written 1 January 1973), *Fibonacci Quart.* **12** (1974), no. 4, p. 335. The letter poses the full series as a conjecture and asserts irrationality after retaining only the distinct running-LCM values.
- [4] T. F. Bloom, *Erdős Problem #269*, erdosproblems.com/269, accessed 28 July 2026 (page displays “last edited 28 December 2025”). The current record labels the finite-support problem open, cites [ErGr80, p. 65] and [Er88c, p. 106], routes to the 1974 letter on p. 335, records the infinite-prime and de-duplicated variants, and explicitly describes its status as the website owner’s present assessment rather than a literature-completeness guarantee.
- [5] The Formal Conjectures Authors, *FormalConjectures.ErdosProblems.269*, Lean source at commit f776d2f, 2025, accessed 28 July 2026.
- [6] T. M. Apostol, *Introduction to Analytic Number Theory*, Springer, New York, 1976.
- [7] A. Hildebrand, *On the number of positive integers $\leq x$ and free of prime factors $> y$* , *J. Number Theory* **22** (1986), 289–307, DOI.
- [8] H. L. Montgomery and R. C. Vaughan, *The Prime Number Theorem*, in *Multiplicative Number Theory I: Classical Theory*, Cambridge Studies in Advanced Mathematics 97, Cambridge University Press, 2007, pp. 168–198, doi:[10.1017/CBO9780511618314.008](https://doi.org/10.1017/CBO9780511618314.008).
- [9] V. Kovač and T. Tao, *On several irrationality problems for Ahmes series*, *Acta Math. Hungar.* **175** (2025), 572–608, doi:[10.1007/s10474-025-01528-0](https://doi.org/10.1007/s10474-025-01528-0); arXiv:2406.17593, 2024.
- [10] Y. Bugeaud and M. Laurent, *Transcendence and continued fraction expansion of values of Hecke–Mahler series*, *Acta Arith.* **209** (2023), 59–90, doi:[10.4064/aa220323-18-1](https://doi.org/10.4064/aa220323-18-1); authors’ publisher-layout PDF, arXiv:2203.12901v1. Theorem 1.1 is on journal p. 61 (publisher-layout PDF p. 3); the authors note there that its $\rho = 0$ case was already obtained by Loxton and van der Poorten.
- [11] J. H. Loxton and A. J. van der Poorten, *Arithmetic properties of certain functions in several variables III*, *Bull. Austral. Math. Soc.* **16** (1977), 15–47.
- [12] S. Fan, comment on Erdős Problem #269, [erdosproblems.com forum, thread 269](https://erdosproblems.com/forum/thread/269), 26 June 2026. The comment gives the two-channel factorisation, the Hecke–Mahler reduction, and the transcendence conclusion for $|P| = 2$; follow-up comments there note the extension to arbitrary coprime pairs.